

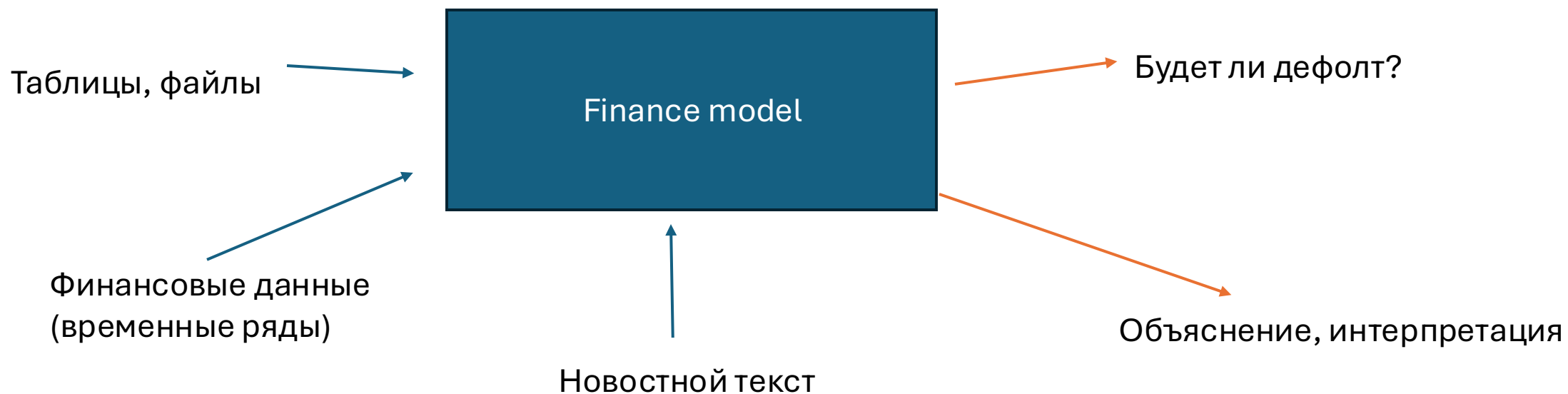
Invertible in meaning text embeddings

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Multimodal models



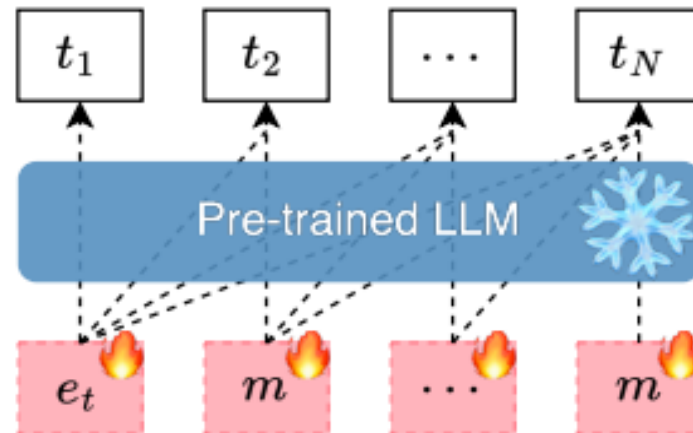
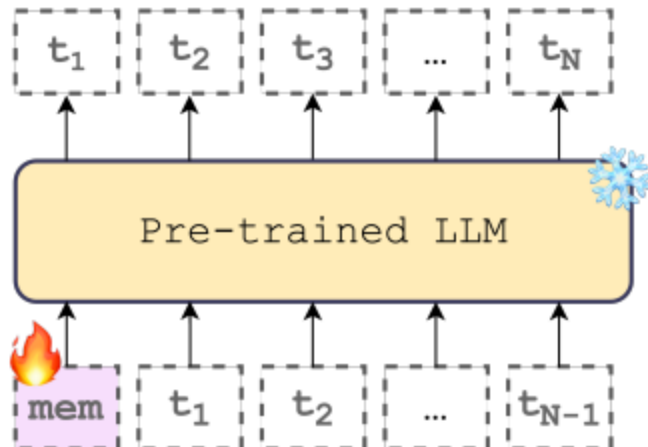
Задача

Проверка гипотезы о возможности (и способах) конструирования такого эмбединга и изучения насколько можно сжимать тексты сохраняя смысл.

Полезные статьи

[1\) Cramming 1568 Tokens into a Single Vector and Back Again: Exploring the Limits of Embedding Space Capacity](#)

[2\) Exploring the Latent Capacity of LLMs for One-Step Text Generation](#)



Financial News Headlines Data

U.S. economy faces significant risks, long road to recovery: IMF staff.

UBS, Morgan Stanley expected to lead Vodafone Tower IPO: sources.

As big U.S. banks let customers delay payments, loan losses remain unclear.

China says it will act to protect its interests after UK Huawei ban.

Метрики

1)Accuracy

2)Bert-score, "microsoft/deberta-xlarge-mnli"

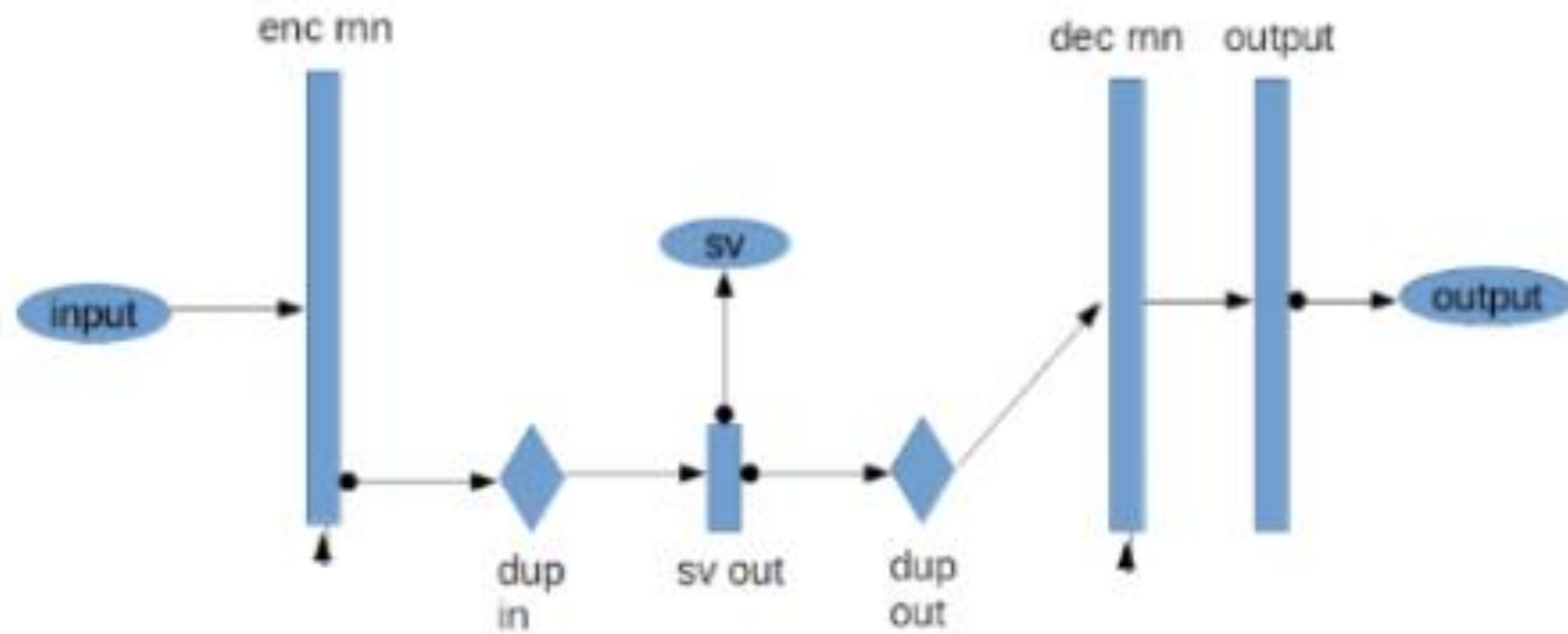
```
score("The car was damaged in the accident.",  
      "The vehicle got wrecked during the crash.") = 0.91
```

```
Score("The economy is growing rapidly.",  
      'My dog loves running in the park.') = 0.35
```

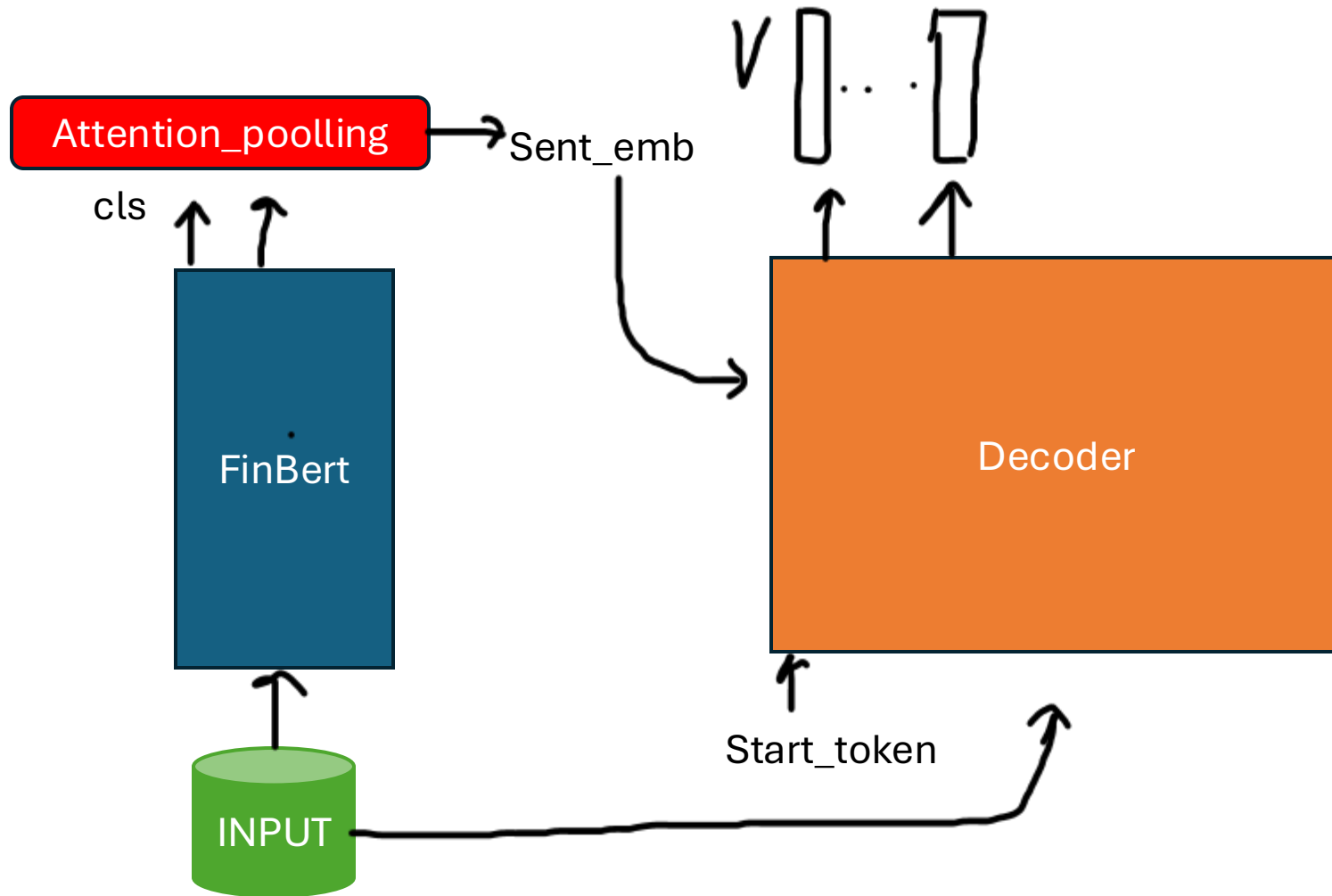
3)paraph_sim, 'sentence-transformers/paraphrase-mpnet-base-v2'

RNN

[Return of the RNN: Residual Recurrent Networks for Invertible Sentence Embeddings](#)



Attention-архитектуры



Обучение

Loss: cross_entropy

optimizer: AdamW

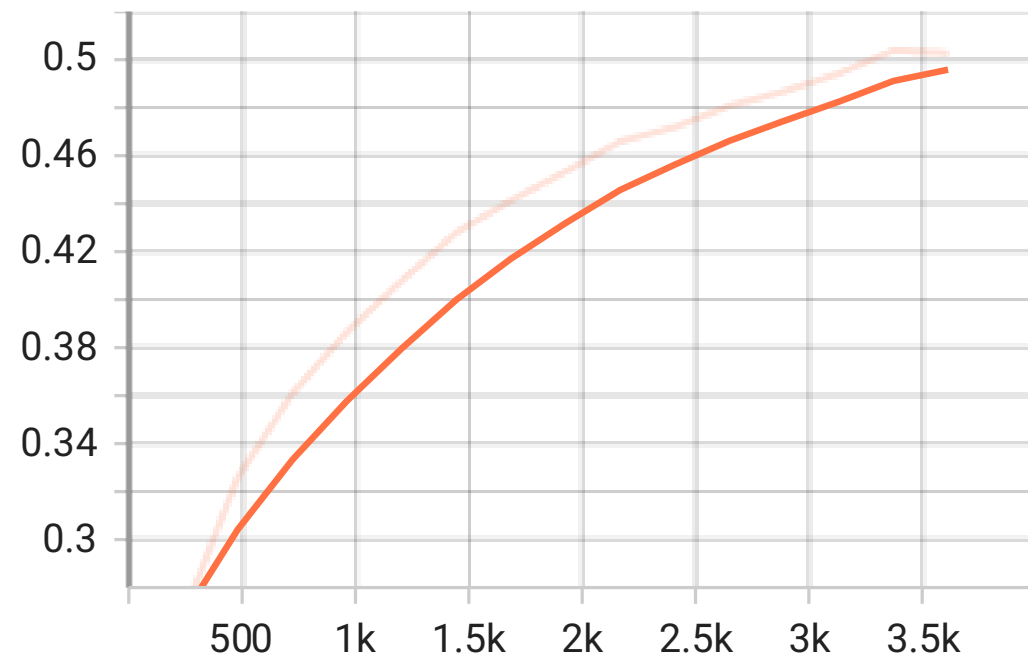
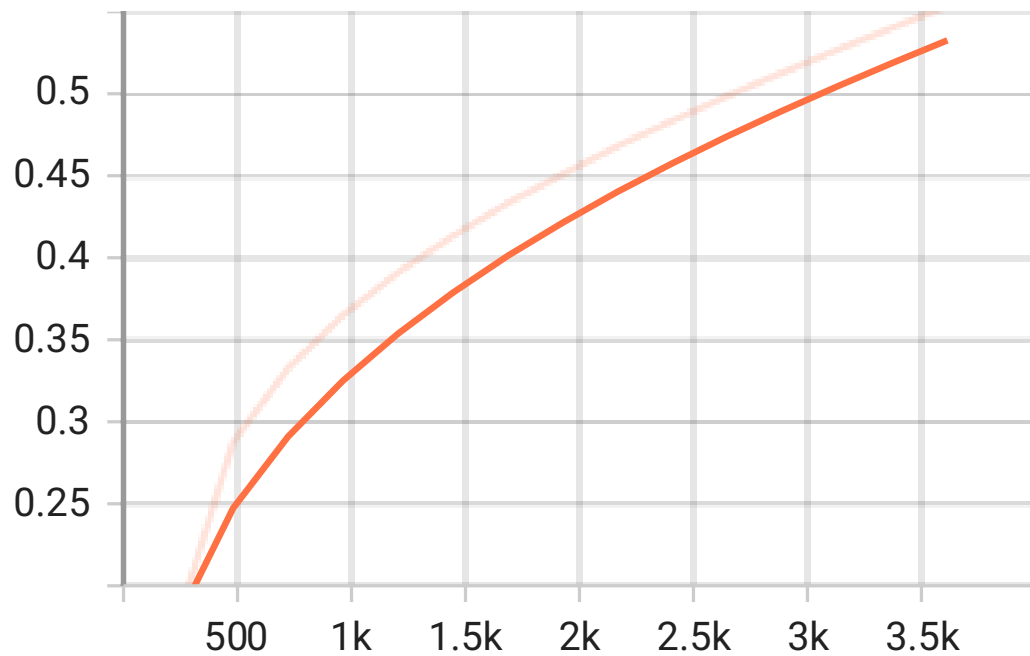
lr_scheduler: ReduceLROnPlateau

batch_size: 128-200

dim	trainable params	non-trainable params
768	38.5 M	109 M
2560	214 M	109 M

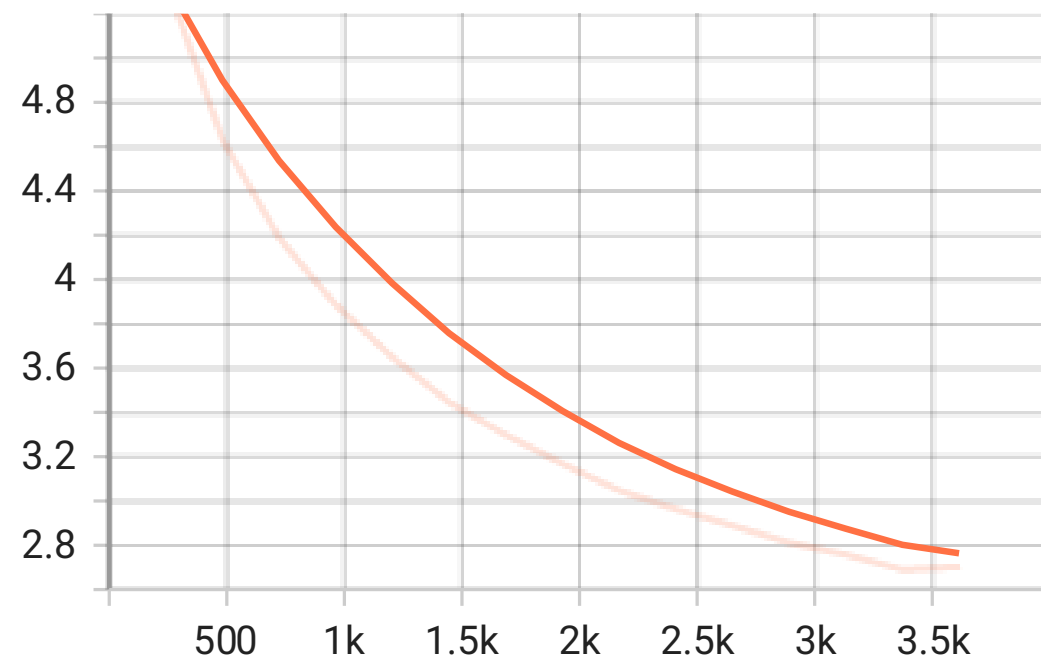
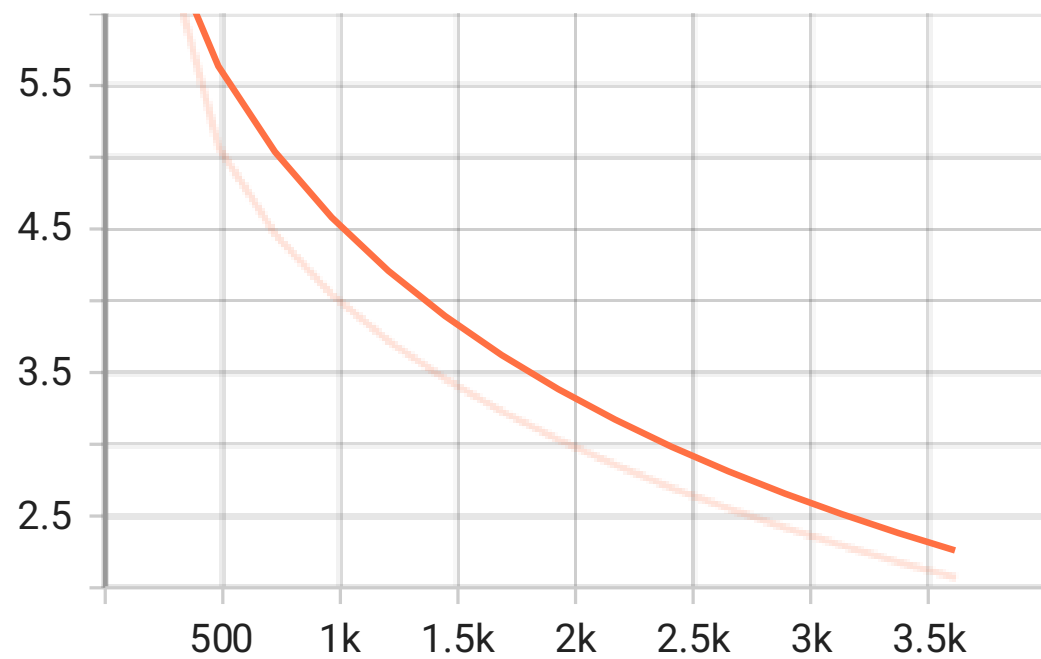
Графики

Accuracy



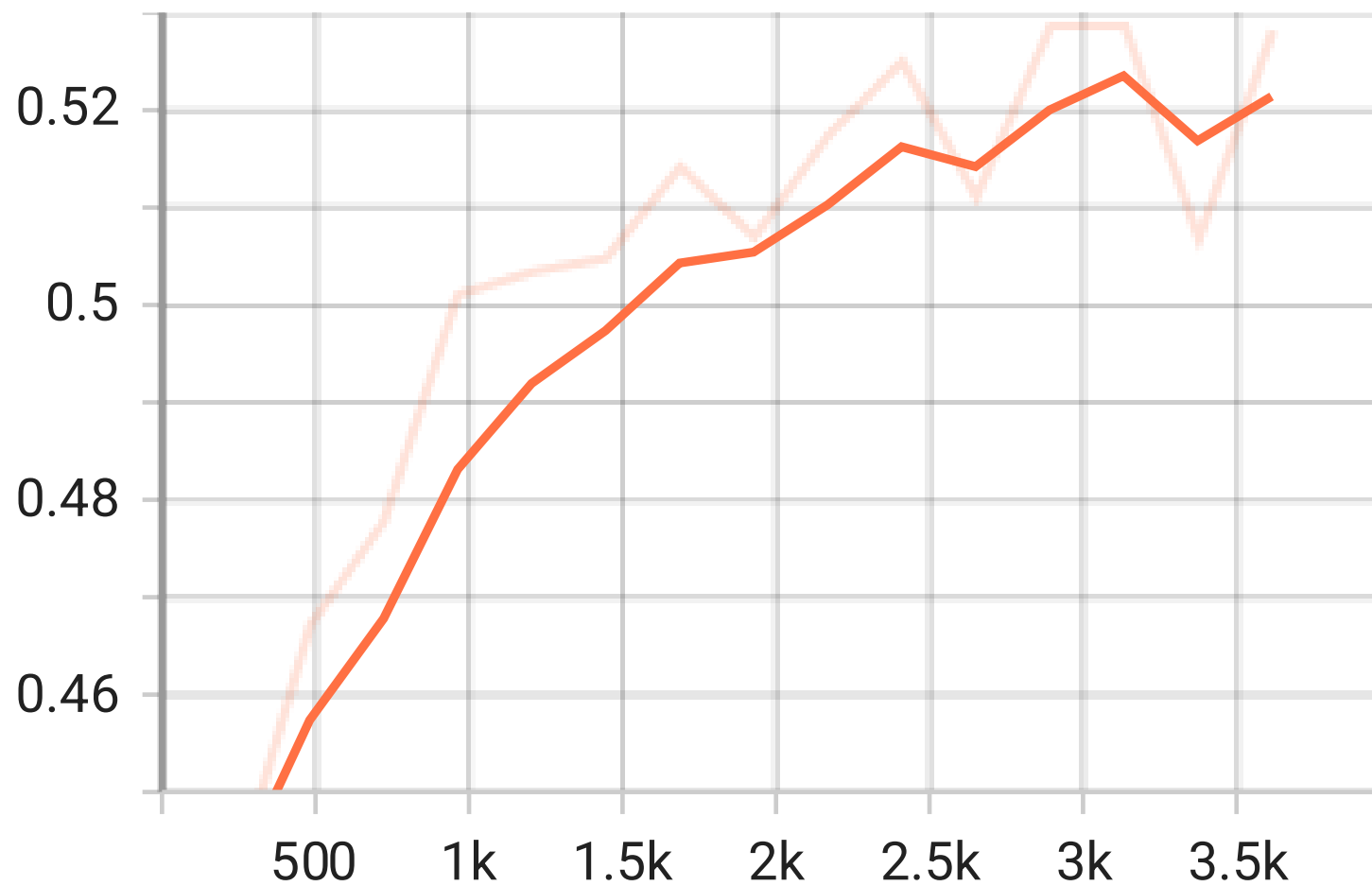
Графики

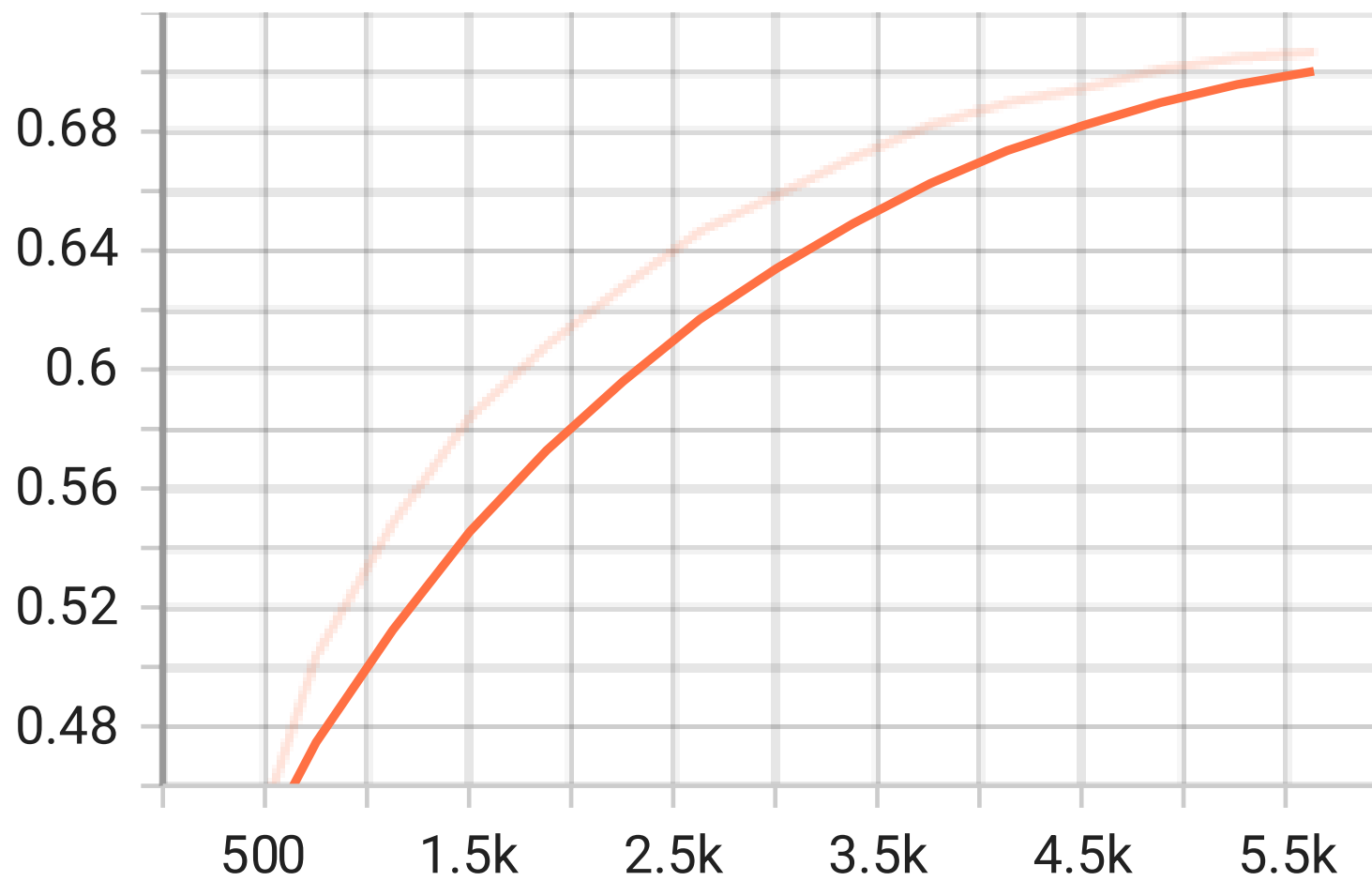
Loss



Графики

Bert-score





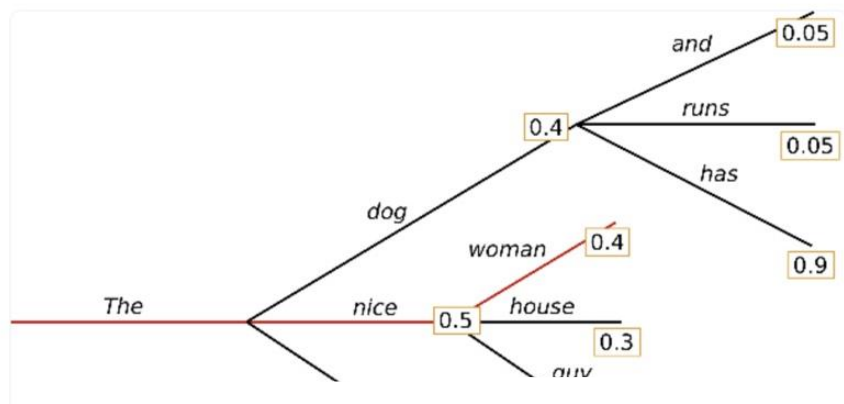
Different dims

model	dim	Valid_accuracy	Bert_score
encoder_only	1568	0.357	
cls_decoder	768	0.406	0.58
att_pool_decoder	768	0.502	0.65
	2560	0.552	0.675

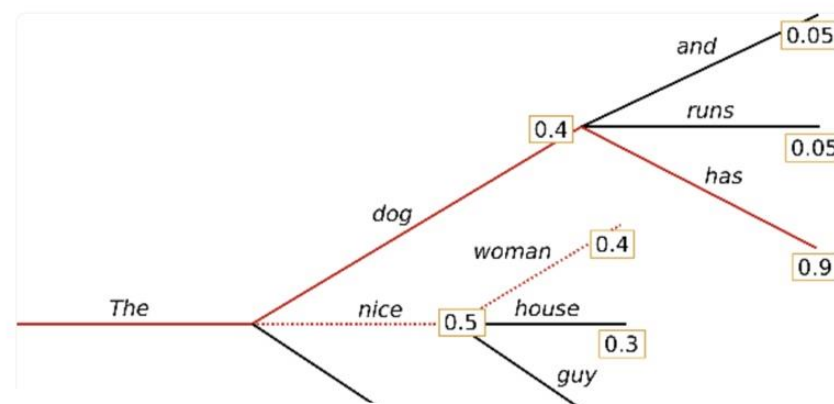
Top-k

Top-k	1	3	5	10	15	25	50	100
Model_768	0.50	0.63	0.68	0.73	0.75	0.78	0.81	0.84
Model-2560	0.55	0.68	0.72	0.77	0.79	0.81	0.84	0.86

Greedy



Beam



beam_width

1

5

10

paraph_sim

0.665

0.687

0.690

cosine_similarity loss

Сэмплируем из logits с помощью gumbel trick.

```
y = F.gumbel_softmax(logits, tau=1.0, hard=True)
```

Straight-Through **Estimator (STE)** $Y = \text{one_hot} + (\text{soft} - \text{soft.detach}())$

```
gen = torch.matmul(y, emb_matrix)
```

paraph_model - paraphrase-mpnet-base-v2

```
Loss = cosine_similarity(paraph_model(input_text), paraph_model(inputs_embeds=gen))
```

Подаем вместо токенов эмбединги токенов gen.

Cosine_similarity loss

Пример похожих по смыслу и нет.

I enjoy reading books in my spare time.

I like reading novels during my free time.

Python is a popular programming language.

paraph_model	my_model
0.918	0.792
0.162	0.544

Loss на перефразах.

Создаем датасет перефраз для train.

3 перефразы на предложение.

gen_paraphrase_model - "ramsrigouthamg/t5-large-paraphraser-diverse-high-quality"

Параметры генерации: top_k=120, top_p=0.9, temperature=1.7,

Loss на перефраззах.

base	paraphrase1	paraphrase2	paraphrase3
Weak sales growth hits American Eagle outlook; shares slide	The American Eagle's outlook is marred by poor sales growth; the stock plummets.	The American Eagle's prospects are harmed by poor sales growth; the stock plummets.	The American Eagle's prospects suffer as a result of poor sales growth; the stock plummets.

ce vs ce_with_paraphrases

- $(\text{ce_loss} + \text{ce_loss1} + \text{ce_loss2} + \text{ce_loss3}) / 4$
- $\min(\text{ce_loss1}, \text{ce_loss2}, \text{ce_loss3})$
- $\text{f1}(\text{ce_loss}, \text{ce_loss1}, \text{ce_loss2}, \text{ce_loss3})$
- Top-5
 - 0.77
 - 0.78

Изучение эмбедингов

Добавить к эмбедингам шум, замерить качество восстановления.

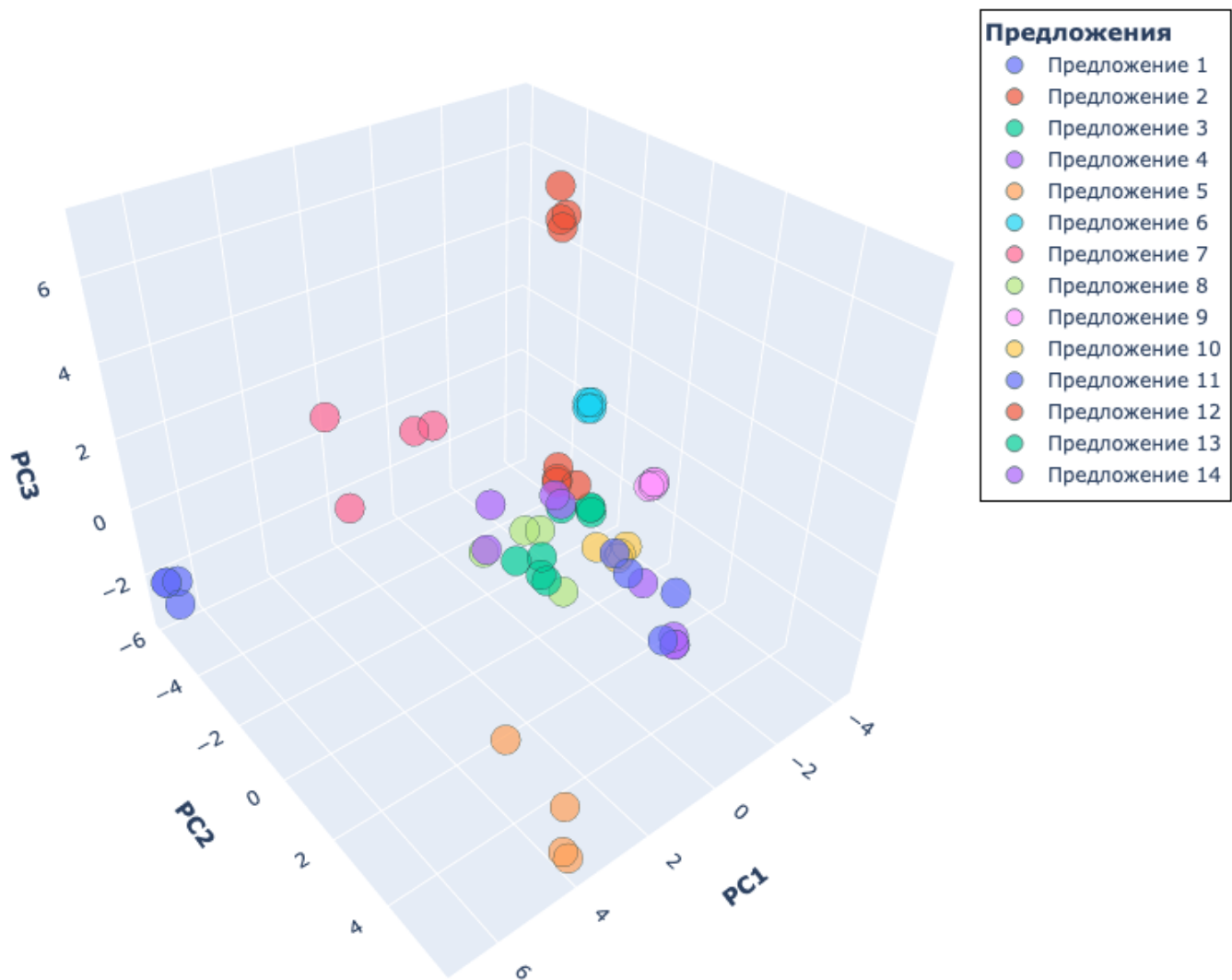
$M=0$

$\text{Bert_score}(\text{sigma} = 0.1) \sim \text{Bert_score}(\text{sigma} = 0)$

$\text{Bert_score}(\text{sigma} = 1) < \text{Bert_score}(\text{sigma} = 0) * 0.9$

3D Визуализация эмбедингов предложений

Наведите/нажмите на точку для просмотра текста перефразы



Примеры генерации

- nils pratley on finance george osborne must understand the imf role is no popularity contest
nils pratley on finance george osborne must think the imf is no need to comment
- oil prices slide as u. s. - china trade war escalates
oil prices slide as u. s. - china trade war escalates
- malaysia regulator to probe if airasia broke rules in airbus deals
malaysia regulator to probe airasia in regulators if boeing deals

Новостные описания

val_accuracy=0.568, paraph_sim=0.595

- Amazon. com is interested in buying prepaid cellphone wireless service Mobile from U. S. carriers - Mobile US and Sprint Corp, two sources familiar with the matter said on.',
-
- 'Amazon. com is interested in buying wireless carriers from U. S. wireless carriers - Mobile US and Sprint Corp, two sources familiar with the matter said on.'
- U. S. companies ' borrowing to spend on capital investments rose 18 % in September from a year earlier, the Equipment Leasing and Association () said on.",
-
- "U. S. companies ' borrowings and investments rose to a record high on Monday in April, the North American Association () said on.")

Параллельные статьи

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